

# Chenshuo (Aiden) Wang

☎ +1 (612)-845-0796 ✉ [chens001@umn.edu](mailto:chens001@umn.edu) 🔗 [linkedin.com/in/aiden-wang](https://www.linkedin.com/in/aiden-wang) 📄 [github.com/chenshuo-aiden-wang](https://github.com/chenshuo-aiden-wang)

## EDUCATION

---

**University of Minnesota – Twin Cities**

Expected: May 2028

*M.S. Mechanical Engineering*

**University of Minnesota – Twin Cities**

Graduated

*B.S. Mechanical Engineering*

GPA: 3.6/4.0

## EXPERIENCE

---

**AltoAuto** | *PyMAPDL, API, CAE Automation, Surrogate Modeling*

**Founder / CAE Automation Engineer**

Nov 2024 – Present

- Automated end-to-end CAD to FEA pipeline, deployed to FSAE and Solar Car team
- Built CAD automation extracting geometry features and machining parameters via SolidWorks API
- Developed automated meshing workflows with geometry cleanup, defeaturing, and element quality enforcement

**Daikin Industries** | *Thermal Modeling, Transient Analysis, Hardware Validation*

**Modeling Engineer**

Sep 2025 – Jan 2026

- Modeled two-phase liquid cooling loop transient dynamics under variable heat load
- Engineered liquid cooling circuits, sizing pump flow rates across worst-case thermal operating margins
- Developed DOE calibration harness sweeping UA, NTU, and pressure drop across valve operating envelope
- Calibrated R-32 two-phase model against hardware bench data, matching across the full operating range

**Cold Plate Design** | *Cold-Plate Optimization, CFD, Chip-Cooling*

**Simulation Engineer ( IEEE ITherm Student Competition )**

Jan 2026 – Present

- Optimized cold plate thermal resistance and pressure drop for a non-uniform chip power map.
- Performing conjugate heat transfer CFD in Ansys to predict junction temperature, pressure drop, and coolant flow
- Developing analytical python models to validate CFD results

**Thermal-Fluids Research Lab** | *Two-Phase Cooling, DAQ, Chip Assembly*

**Research Assistant**

Jan 2026 – Present

- Investigating phase-change cooling for high-power semiconductors to maximize chip-scale heat flux removal.
- Correlated high-speed visualization of surface bubble dynamics with boiling curve heat transfer performance.
- Assembled chip test articles, wired thermocouple arrays, and built the DAQ pipeline for Critical Heat Flux analysis.

**UMN Solar Vehicle** | *Mechanical Design, Manufacturing, Battery Thermal*

**Mechanical — Battery Thermal**

Nov 2022– May 2025

- Designed the battery enclosure with integrated air intake and exhaust routing in SolidWorks
- Wired and placed NTC thermistors on each cell group; configured BMS thermal cutoff thresholds
- Logged cell temperature data during track testing; identified hot spots and revised vent placement

## SKILLS

---

**Core Mechanical:** Thermodynamics, Fluid Mechanics, Heat Transfer, Solid Mechanics, GD&T (ASME Y14.5), Dynamics & Kinematics, Liquid Cooling Systems, Two-Phase Flow, Thermal Management

**Computational Engineering:** Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Turbulence Modeling, Meshing, Model-Based Design (MBD)

**Mechatronics & Control:** PID Control, Hardware-in-the-Loop (HIL), System ID, Real-Time DAQ, Electromechanical Actuation, MPC, LQR, EKF, Kalman Filter

**Modeling & Simulation:** ANSYS (Mechanical, Fluent, Icepak), Simulink, Simscape, SolidWorks, Creo, OpenFOAM, Code\_Aster, Gmsh, RoboDK, Onshape

**Programming & Scripting:** Python, C/C++, MATLAB, PLC, Git, PyMAPDL, PyFluent, PyVista, SolidWorks API, PyTorch, FORTRAN